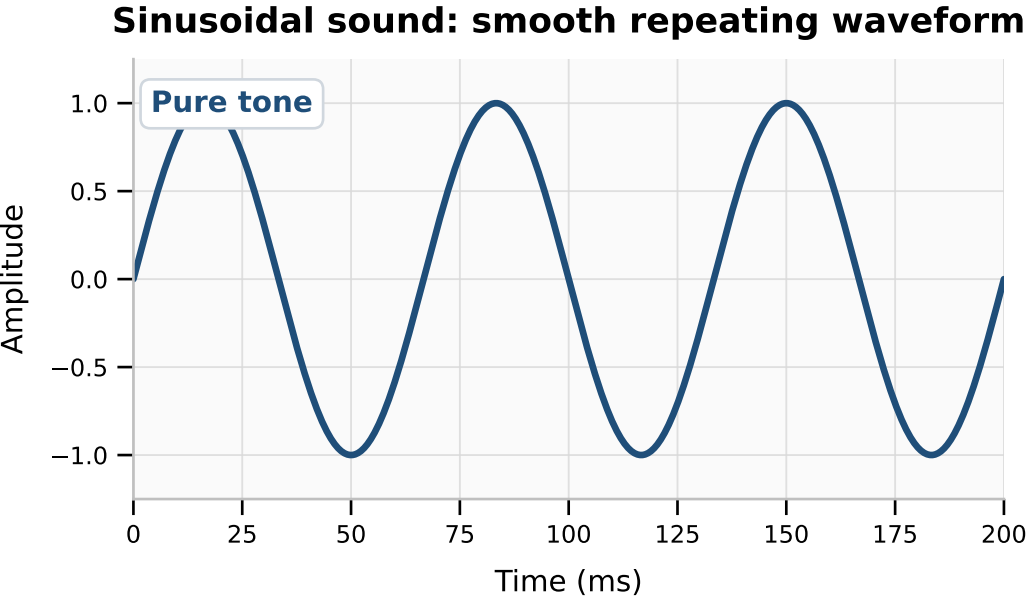
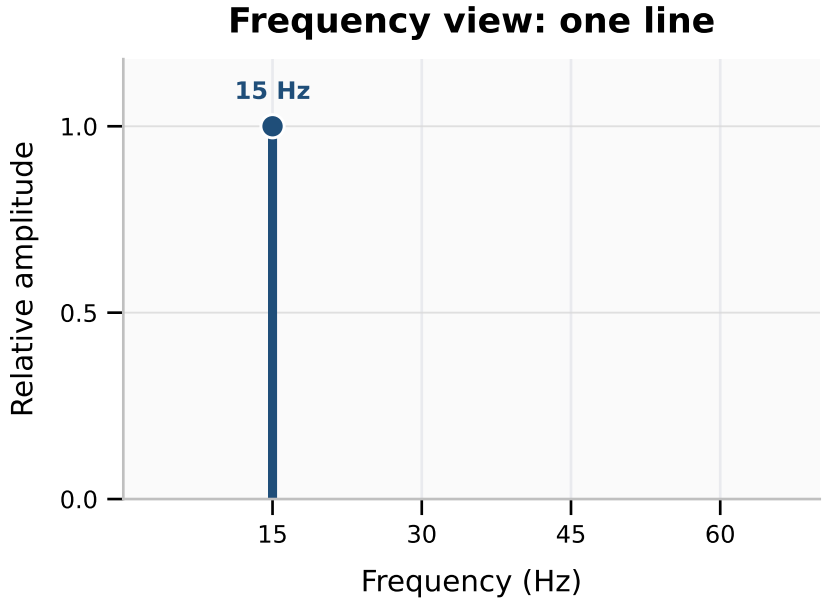
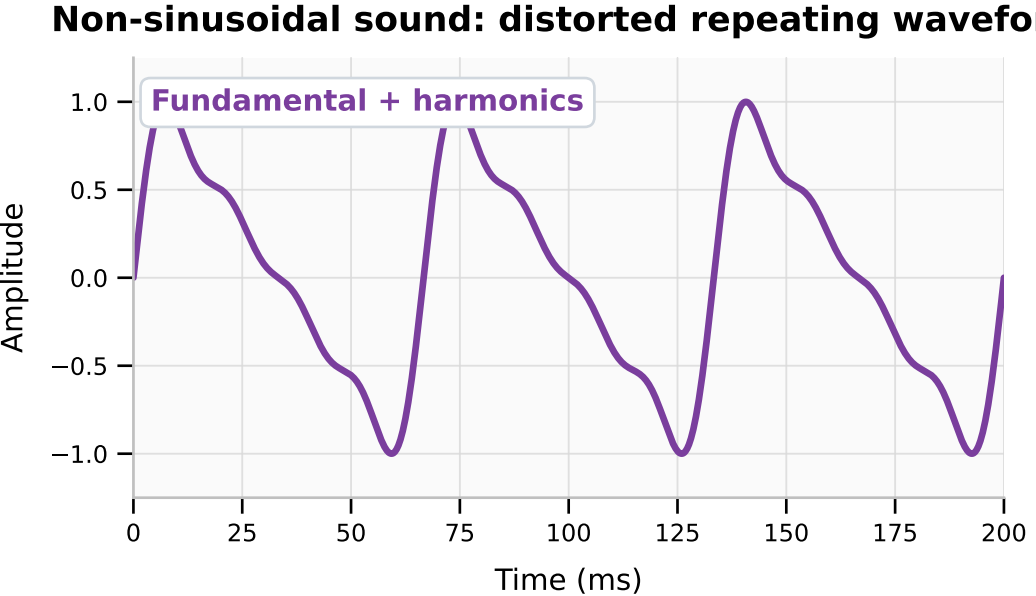
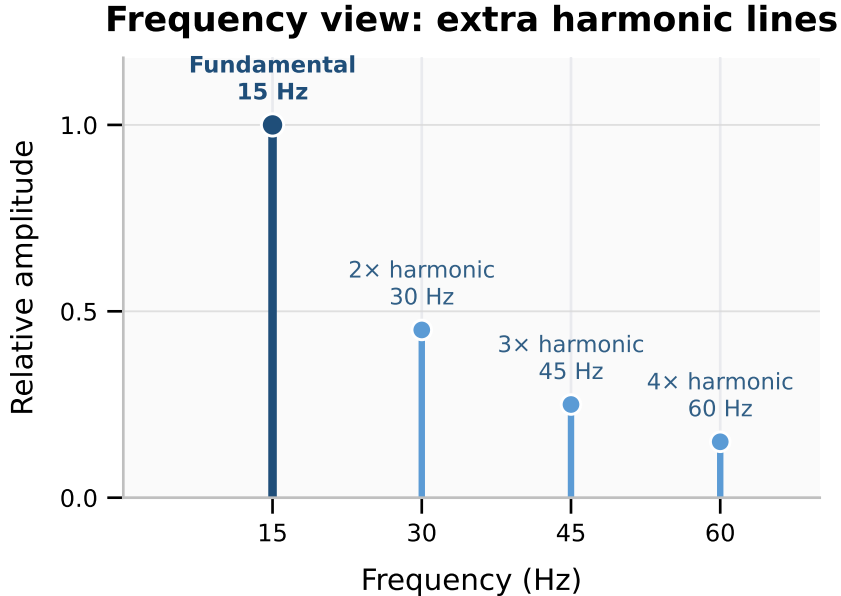
Sinusoidal vs Non-Sinusoidal Sound: Why Harmonics Appear

A pure sine wave produces one frequency line. A distorted repeating waveform produces the fundamental plus harmonics.